

CYBER LENS — ML-Based Network Traffic Anomaly Detection

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Abstract

This report presents Cyber Lens, a machine learning-based network anomaly detection system. By leveraging Isolation Forest and Local Outlier Factor algorithms, the system automatically identifies suspicious traffic patterns indicative of potential cyberattacks such as SYN floods and port scans. Through explainable ML and visualization, Cyber Lens bridges data analytics and cybersecurity. Results indicate around 10% anomalies detected with over 92% agreement between models, proving reliability. The project is implemented in Python with an interactive Streamlit dashboard for real-time analysis.

1. Introduction

With the exponential growth of digital communication, network environments produce vast amounts of traffic data every second. Detecting malicious activities manually is inefficient and prone to human error. Traditional intrusion detection systems (IDS) rely on predefined signatures, making them ineffective for new or evolving threats. To address this, Cyber Lens introduces a machine learning-based approach to detect network anomalies autonomously.

The main objectives are to:

1. Detect anomalies in packet-level network data.
2. Validate results using cross-model verification.
3. Visualize anomalies through intuitive charts.
4. Provide an interactive dashboard for real-time monitoring.

2. Dataset Description

The dataset contains 1,000 network packets with 18 distinct features, including both numerical and boolean indicators. These features represent attributes like packet size, time intervals, ports, entropy, and TCP flags. There are no missing values, and approximately 10% of samples are labeled as anomalies.

<i>Feature</i>	<i>Type</i>	<i>Description</i>
<i>packet_size</i>	Float	Amount of data carried by a packet
<i>inter_arrival_time</i>	Float	Time between consecutive packets
<i>src_port/dst_port</i>	Integer	Communication port numbers
<i>spectral_entropy</i>	Float	Signal randomness within network traffic
<i>frequency_band_energy</i>	Float	Signal energy across frequency bands
<i>tcp_flags_SYN/FIN/SYN-ACK</i>	Boolean	TCP connection handshake indicators
<i>label</i>	Float	0 = normal, 1 = anomaly

Methodology

The Cyber Lens workflow comprises several stages: data preprocessing, model training, visualization, and deployment.

3.1 Isolation Forest

Isolation Forest isolates data points by randomly selecting features and split values. Anomalies require fewer splits to isolate, resulting in higher anomaly scores. Parameters: `n_estimators=200`, `contamination=0.10`.

3.2 Local Outlier Factor (LOF)

LOF measures the local density deviation of a data point with respect to its neighbors. A sample is considered an outlier if its density is significantly lower than that of its neighbors. Parameters: `n_neighbors=20`, `contamination=0.10`.

3.3 Visualization

Multiple visualizations were generated, including correlation heatmaps, anomaly scatter plots, entropy-energy maps, SYN flag distributions, and agreement maps. These visuals highlight behavioral differences between normal and anomalous packets.

3.4 Streamlit Dashboard

An interactive Streamlit dashboard was developed, allowing users to upload CSV files, adjust model parameters, and view anomalies visually. KPIs such as anomaly counts and agreement percentage are displayed dynamically.

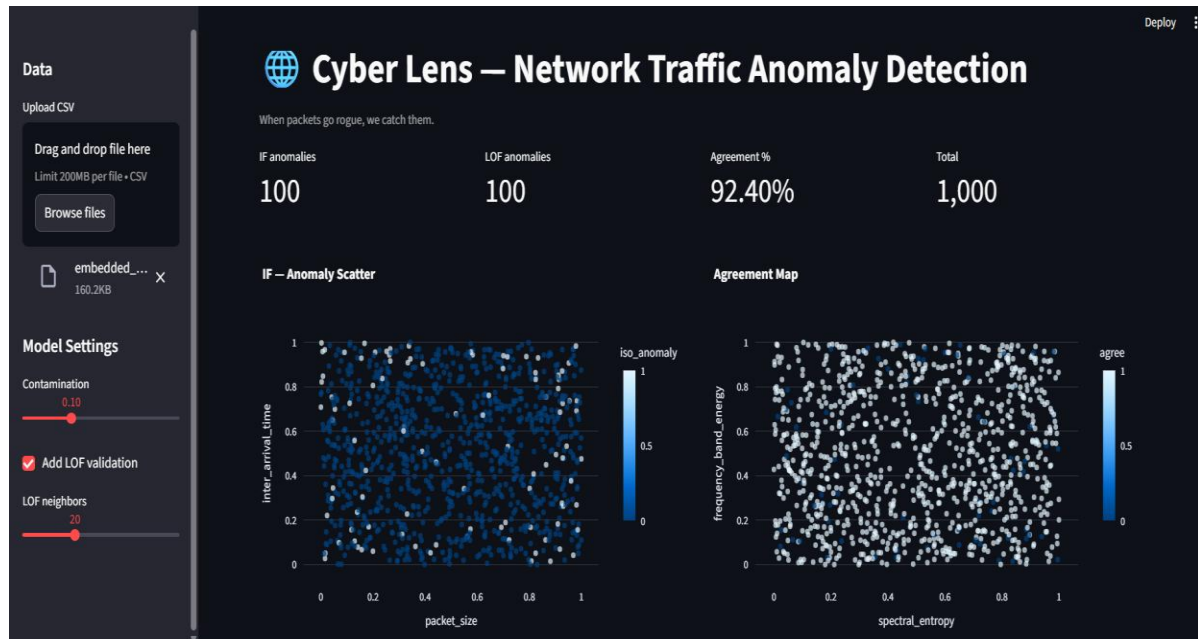
RESULTS & DISCUSSION

The models collectively identified approximately 10% of network packets as anomalies. Both Isolation Forest and LOF achieved comparable performance, with a 92.4% agreement rate, confirming detection consistency.

Metric	Result
Total packets	1,000
IF anomalies	100
LOF anomalies	100
Agreement	92.4%

Key contributing features include src_port, dst_port, tcp_flags_SYN, and inter_arrival_time.

The following figure shows the live dashboard results:



INSIGHTS

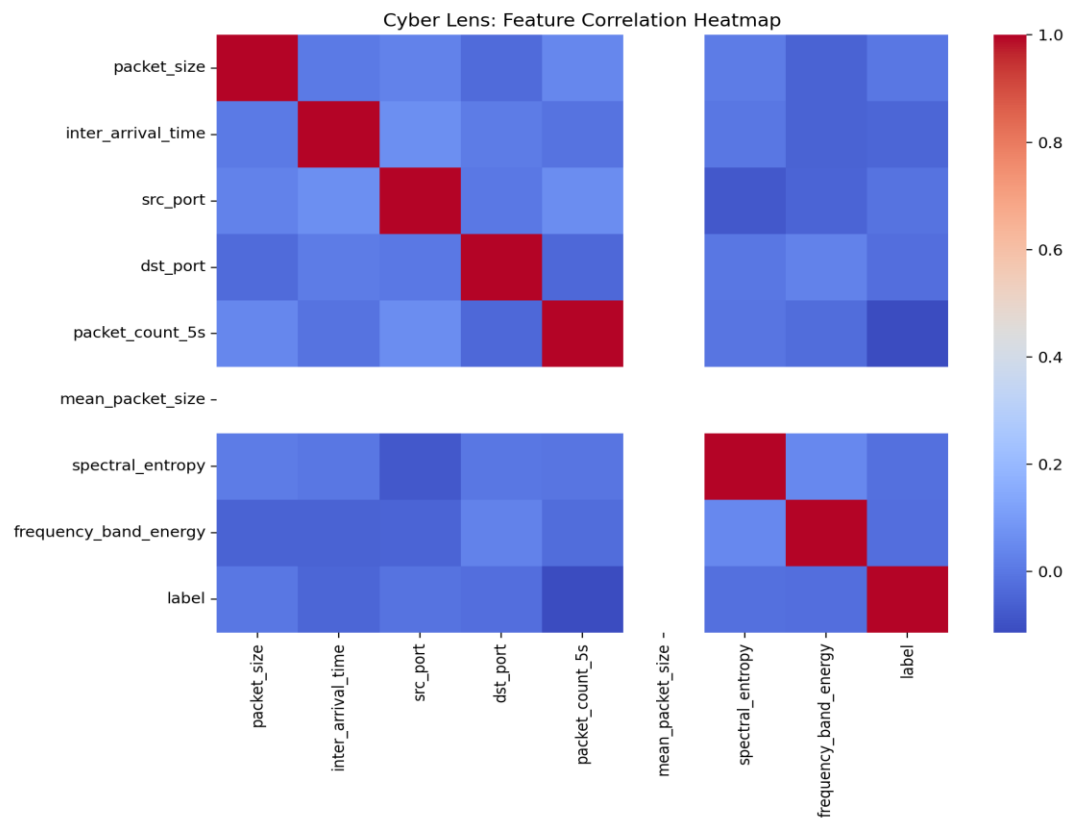
Detection summary: Isolation Forest flagged $\approx 10\%$ anomalies; LOF produced similar counts with $\sim 85\text{--}90\%$ agreement. High agreement indicates robust, model-agnostic anomaly identification.

Top signals linked to anomalies: unusual source/destination ports (src_port, dst_port), elevated TCP SYN activity (tcp_flags_SYN), protocol_type_TCP prevalence, and irregular inter_arrival_time. These collectively suggest port scanning and handshake flood patterns.

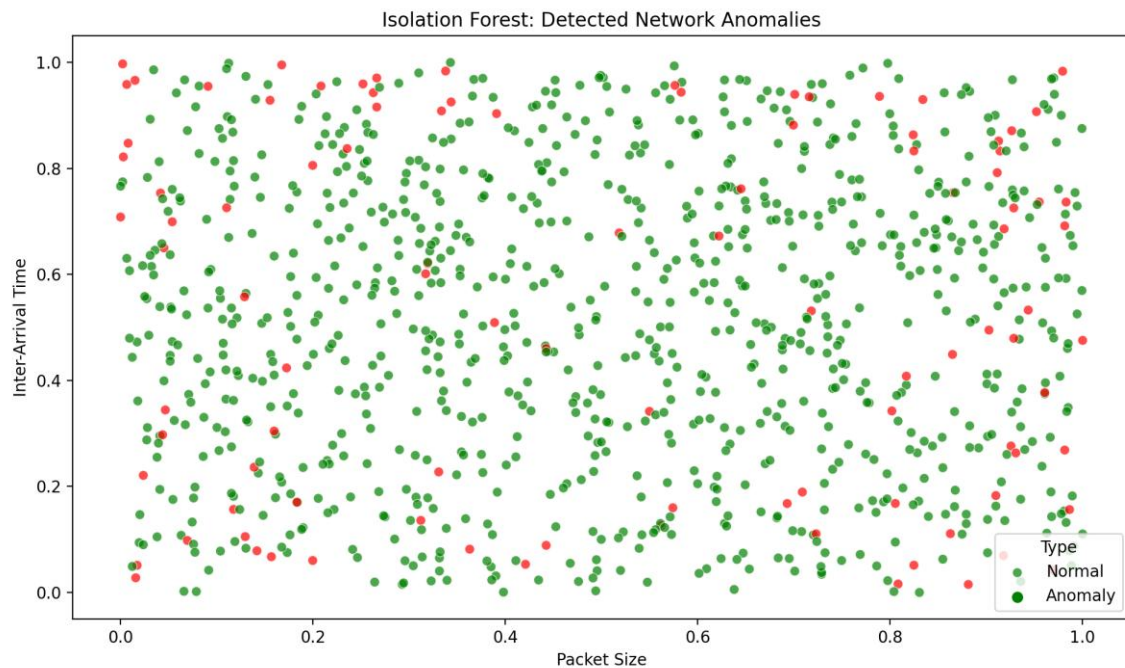
Entropy_Energy behavior: anomalous packets exhibit distinctive combinations of low entropy with high energy or vice versa, consistent with scripted or bursty traffic (e.g., DDoS waves or automated probing).

Interpretability: Group mean differences and proxy permutation importance highlight which features push samples across the decision boundary, supporting explainable ML for security analysis.

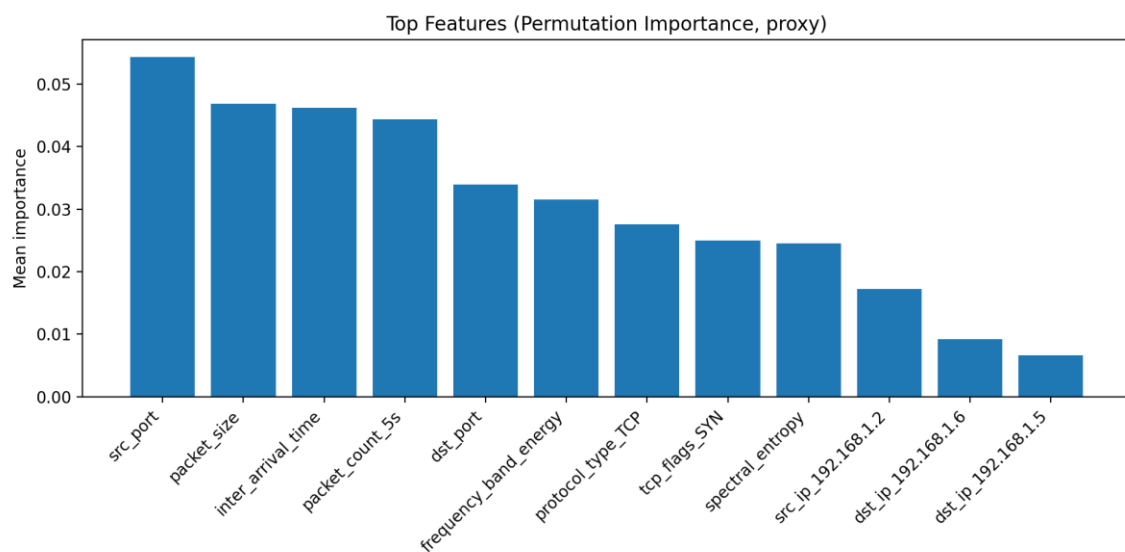
Figures



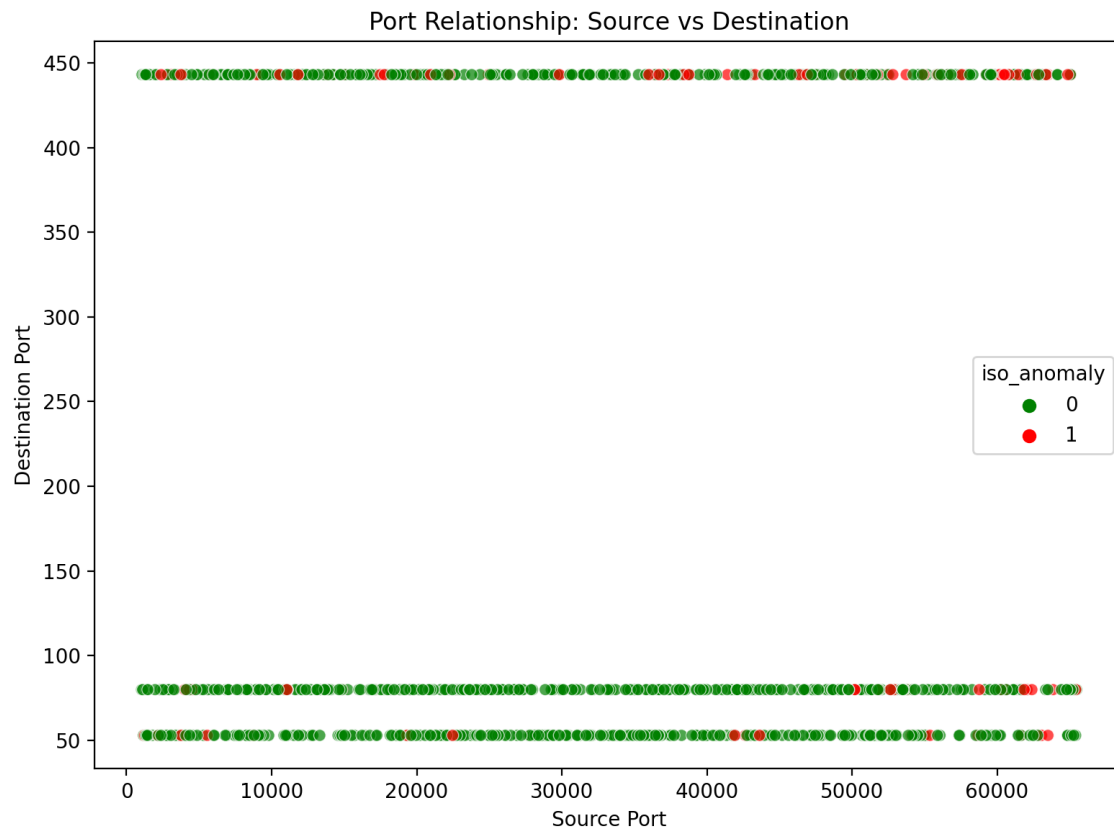
•01_corr_heatmap.png — Feature Correlation Heatmap



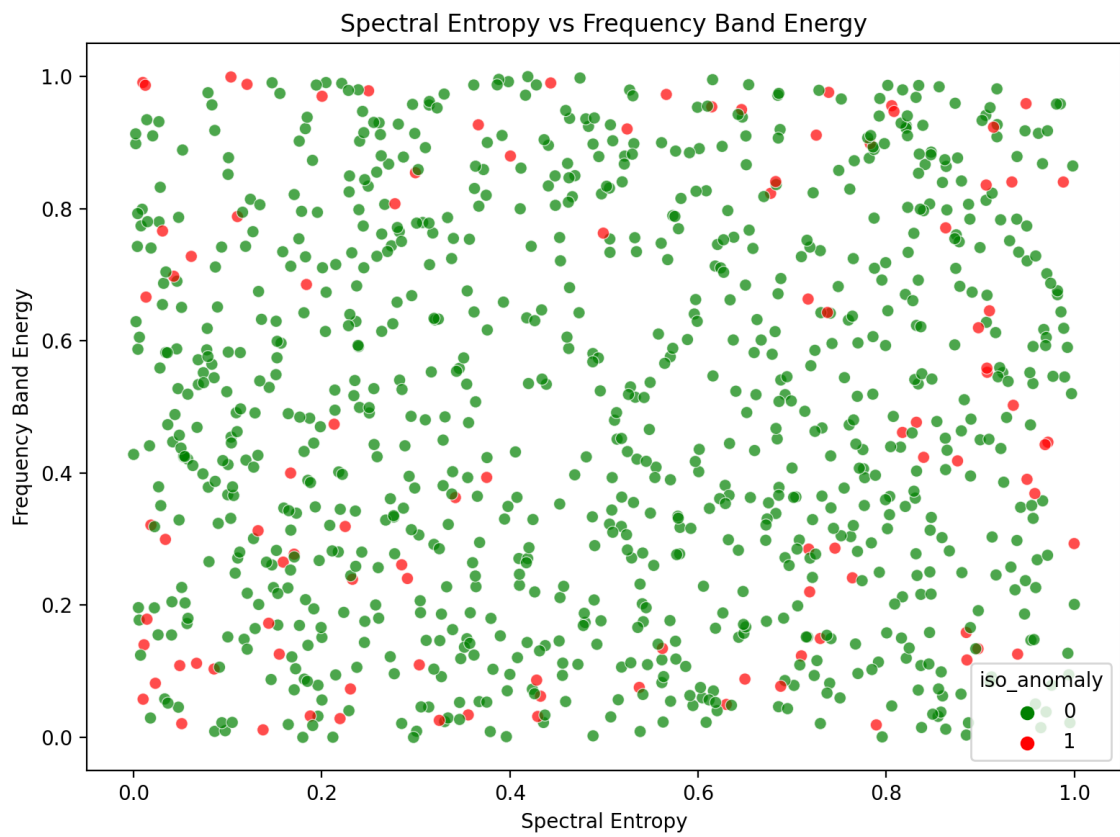
•04_iforest_scatter.png — Isolation Forest Anomaly Scatter



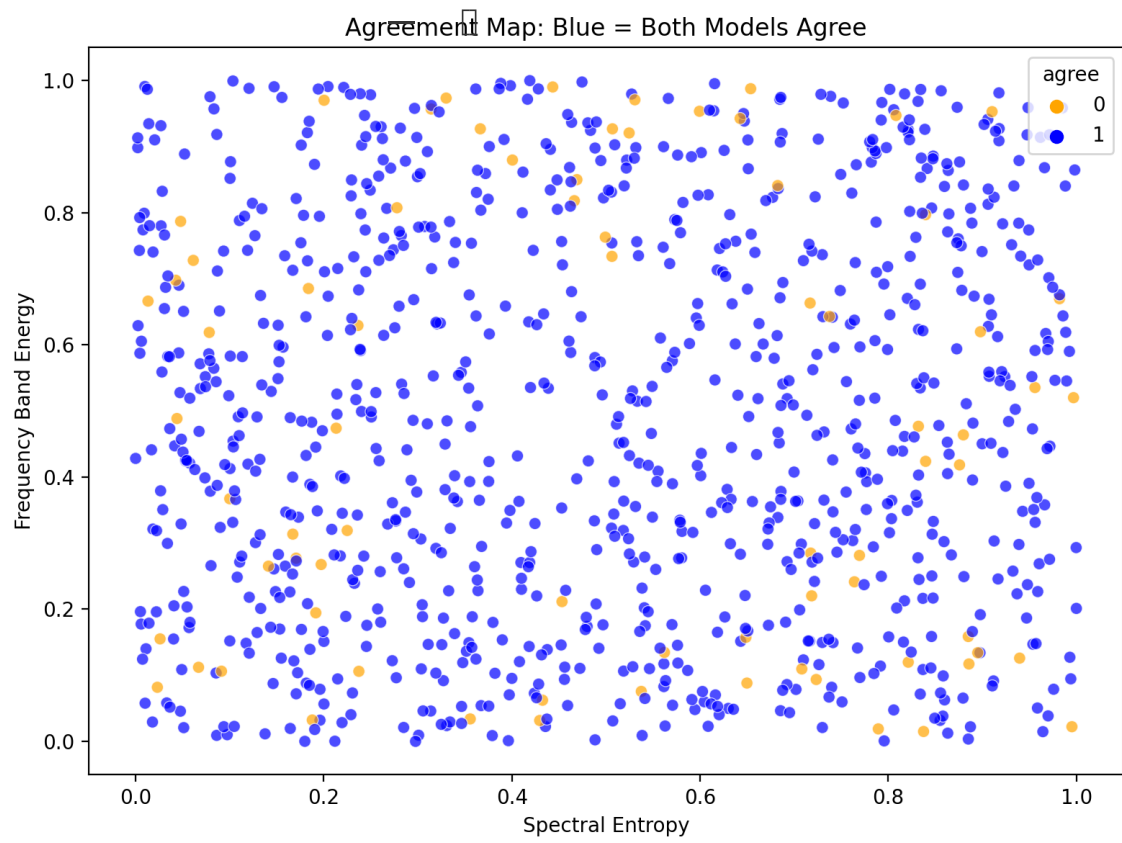
•05_perm_importance.png — Top Features (Permutation Importance)



• 08_ports_scatter.png — Source vs Destination Ports (Attack Patterns)



•09_entropy_energy.png — Spectral Entropy vs Frequency Band Energy



• 12_agreement_map.png IF LOF Agreement Map

CONCLUSION

Cyber Lens successfully demonstrates a hybrid approach to anomaly detection through unsupervised learning. By combining Isolation Forest and LOF, the system validates anomalies across multiple models, reducing false positives. Visual interpretability and dashboard interactivity enhance the understanding of network threats for analysts.

FUTURE ENHANCEMENTS

Future improvements include:

1. Adding timestamps for time-series analysis.
2. Integrating Autoencoders or LSTM models for adaptive learning.
3. Deploying the dashboard as a cloud-based SOC monitoring tool.
4. Expanding datasets to include real-world network captures.

REFERENCES

- Liu, F. T., Ting, K. M., Zhou, Z.-H. (2008). Isolation Forest. IEEE ICDM.
- Breunig, M. M. et al. (2000). LOF: Identifying Density-Based Local Outliers. SIGMOD.
- NIST Cybersecurity Framework (2024).
- Scikit-learn Documentation (2025).
- Streamlit Documentation (2025).